Key Competencies & Knowledge Areas

- **Programming Languages:** Python (primary), SQL (for data manipulation), basics of R (optional)
- **Data Manipulation & Cleaning:** pandas, numpy, SQL, Excel
- **Data Visualization:** matplotlib, seaborn, Power BI, Tableau
- **Statistical Analysis:** Descriptive & inferential stats, hypothesis testing
- **Data Engineering:** ETL processes, database management, working with APIs, basics of cloud data tools
- **Business Intelligence & Dashboarding:** Power BI, Tableau, Google Data Studio
- **Basic Machine Learning:** scikit-learn for simple models
- **Fullstack Integration:** Connecting front-end interfaces (dashboards/web apps) to backend data sources
- **Soft Skills:** Data storytelling, communication, project documentation

Sequenced Learning Milestones

Short-Term (Months 1–3): Foundational Skills

1. **Learn Python for Analytics**
    - **Resources:** Automate the Boring Stuff, Kaggle Python Track
    - **Project:** Analyze a real dataset (e.g., sales data, COVID-19 data) and summarize key findings in a notebook.
2. **Master SQL & Data Manipulation**
    - **Resources:** SQL for Data Science (Coursera), Mode SQL Tutorial
    - **Project:** Create and query a database of business transactions.
3. **Intro to Data Cleaning & Preparation**
    - **Resources:** Kaggle Pandas Track, Data Cleaning in Python (DataCamp)
    - **Project:** Clean and prepare a messy dataset for analysis.

Mid-Term (Months 4–7): Visualization, Engineering & BI

4. **Data Visualization & Storytelling**
   - **Resources:** Storytelling with Data (Book), Tableau Free Training, Seaborn Docs
   - **Project:** Build comparative dashboards visualizing trends and anomalies.
5. **Learning Business Intelligence Tools**
   - **Resources:** Power BI Guided Learning, Google Data Studio
   - **Project:** Develop an interactive BI dashboard for business leaders.
6. **Data Engineering Foundations**
   - **Resources:** Intro to Data Engineering (DataCamp), ETL Concepts (Medium)
   - **Project:** Setup an ETL pipeline using Python to automate data ingestion and cleaning.

Long-Term (Months 8–12): Fullstack Integration & Advanced Analytics

7. **Working with APIs and Web Data**
   - **Resources:** Python Requests Docs, Real Python: API Consumption
   - **Project:** An app that pulls data from an external API and visualizes/report insights.
8. **Connecting Analytics to Apps/Dashboards**
   - **Resources:** Flask Mega-Tutorial (Web App Basics), Streamlit Docs
   - **Project:** Build a simple web dashboard (Flask/Streamlit) connected to a database or live data source.
9. **Advanced Data Analytics & Simple Machine Learning Models**
   - **Resources:** Kaggle Machine Learning Track, scikit-learn User Guide
   - **Project:** Analyze a business dataset, build a simple ML model for prediction, and integrate results into visualization/dashboard.
10. **Portfolio & Communication**

- **Resources:** Create a Github Portfolio
- **Project:** Publish and showcase each milestone project; prepare "case studies" explaining your process and results for resumes or job-seeking.

Step-by-Step Structure

Step 1: Gain Python & SQL Mastery

- Take a beginner course (Kaggle, DataCamp, Coursera).
- Practice coding daily (30 min–1 hour).
- Project: Clean, analyze, and visualize a public dataset.

Step 2: Explore Visualization & BI Tools

- Learn Tableau/Power BI basics through guided online lessons.
- Build interactive dashboards; share via LinkedIn/GitHub.

Step 3: Learn Data Engineering & ETL Basics

- Create an ETL pipeline automating the collection, cleaning, and storage of data.
- Experiment with cloud tools if possible (bigquery, AWS basics).

Step 4: Build Fullstack Data Apps

- Learn how to display insights and visualizations in a web app (Flask/Streamlit).
- Integrate backend analytics with frontend dashboards.

Step 5: Apply Advanced Analytics & ML Basics

- Build a simple predictive model (scikit-learn).
- Integrate model outputs into business dashboards.

Step 6: Develop Portfolio & Professional Presence

- Document projects with clear READMEs.

- Create blog posts/videos explaining key findings and dashboards.
- Update Linkedin and GitHub monthly.

Short-Term Goals (Months 1-6):

- Complete 3+ small projects (one each for manipulation, visualization, ETL).
- Master Python, SQL, and one BI tool.
- Build and share at least one dashboard.

Long-Term Goals (Months 7-12):

- Build a fullstack analytics web app.
- Automate a regular job-related task with an ETL/script solution.
- Create a portfolio with at least three major projects.
- Prepare for an entry-level or mid-level fullstack analytics interview.

Progress Tracking:

- **Weekly:** Track learning hours, course completion, and skills gained.
- **Monthly:** Complete/refine mini-projects.
- **Quarterly:** Publicly share work, gather feedback, and adjust goals.

Following this roadmap will turn foundational skills into advanced, practical fullstack analytics capabilities, ready for business impact and future career growth.